Module 0: Prerequisite (Indexing and Searching Algorithms)

#	Question	Marks	Month & Year
1	Apply Boyer Moore algorithm to construct bad match table and find the index of the given pattern for the string below with steps. String: STUDENTS ARE SMART I LIKE SMART STUDENTS; Pattern: SMART	10	May 2025

Module I: Introduction

#	Question	Marks	Month & Year
1	Discuss Objective of IRS	5	Dec 2024
2	Discuss Objective of IRS	5	May 2024
3	Recall objectives of Information Retrieval Process	5	May 2025
4	Differentiate between Data Retrieval and Information Retrieval	5	Dec 2025
5	Differentiate between Information versus Data Retrieval	5	May 2025
6	Explain the Retrieval Process with neat labelled diagram	5	Dec 2025
7	List and define different parameters to judge the quality of information	5	May 2023
8	Different types of Information Systems (short note)	10	May 2025

Module II: IR Models

#	Question	Marks	Month & Year
1	Explain the taxonomy of Information retrieval model with classification diagram	10	Dec 2023
2	Show taxonomy of IR models and recall browsing models in detail	10	May 2025
3	Compare boolean model and vector model	5	Dec 2023
4	Explain Classic Information Retrieval models	5	May 2023
5	Explain Boolean Models in detail with example	10	May 2025
6	What is the significance of tf and idf? How can you calculate tf and idf in a vector model?	10	Dec 2023
7	Explain ranking and similarity measures with suitable example	10	Dec 2023
8	Vector model ranking problem (D1: I went to the park to play; D2: park is nearby to play; D3: going to the park is fun; Q: park nearby play)	10	Dec 2024
9	Vector model ranking problem (D1: I went to the park to play; D2: park is nearby to play; D3: going to the park is fun; D4: park nearby play)	10	Dec 2025
10	Vector space ranking problem (d1: beautiful garden; d2: evening garden time; d3: garden time is beautiful; Q: garden time) using tf-idf and cosine similarity	10	May 2025

#	Question	Marks	Month & Year
11	Explain Probabilistic Retrieval Model. Compare and contrast Belief Network Model and Inference Network Model	10	Dec 2024
12	Explain the process of Structured Text retrieval model	5	Dec 2024
13	What is Structured text retrieval Models? Explain different types with example	10	Dec 2025
14	Explain Object Oriented and Object Relational DBMS	10	May 2023
15	What is flat browsing and hypertext browsing? Explain	10	Dec 2023
16	Write different issues for modeling natural language and find solution for it	10	Dec 2024
17	Explain Natural language modelling issues and its solution	10	May 2025
18	Summarize different issues for modeling natural language (short note)	10	Dec 2025
19	Define Natural Language with features and applications	5	May 2023

Module III: Query Processing and Operations

#	Question	Marks	Month & Year
1	Illustrate different types of keyword-based queries	5	Dec 2023
2	Illustrate different types of keyword-based queries	5	Dec 2024
3	What is the purpose of using keyword-based query? Briefly explain any 3 types of keyword-based queries	10	Dec 2025
4	What is the purpose of using keyword based query? Briefly explain any 3 types of keyword based queries	10	May 2024
5	What is the purpose of using keyword based query? Briefly explain any 3 types of keyword based queries	10	May 2025
6	State the different types of queries. Explain the pattern matching query concept with an example	10	Dec 2023
7	Compare and contrast Proximity Queries and Wildcard Queries	5	May 2024
8	Explain Relevance Feedback. Discuss the contribution of relevance feedback for quality IR	10	May 2023
9	What is User relevance feedback: How this will contribute to get results	10	May 2024
10	User relevance feedback (short note)	5	Dec 2023
11	User relevance feedback (short note)	5	Dec 2024
12	How user relevance feedback model can be applied to improve search results in search recommendation engine (short note)	10	Dec 2025
13	Discuss automatic local analysis, with suitable example	10	May 2023
14	Discuss automatic local analysis, with suitable examples	10	May 2024
15	Discuss Multimedia IR Model	10	May 2023
16	Define data modeling? Explain any two types of data models	5	May 2024

Module IV: Text Processing

#	Question	Marks	Month & Year
1	Illustrate different markup languages with its applications	5	Dec 2024
2	Illustrate any 2 markup languages and its applications	10	Dec 2025
3	Different markup languages and its applications (short note)	10	May 2025
4	What is Metadata? Explain Descriptive and Semantic Metadata in detail	10	Dec 2025
5	Importance of metadata and its types	5	May 2025
6	Discuss various phases of text preprocessing within a document. Discuss any one application for the same	10	Dec 2024
7	Discuss various phases of text preprocessing within a document. Discuss any one application for the same	10	May 2024
8	Lexical analysis (short note)	5	May 2023
9	Write a note on Scalar and Metrics Clusters	10	May 2023
10	Document clustering (short note)	10	May 2024
11	Information Retrieval in digital libraries (short note)	5	Dec 2023
12	Information Retrieval in digital libraries (short note)	5	Dec 2024

Module V: Indexing and Searching

#	Question	Marks	Month & Year
1	Explain inverted file indexing with suitable examples	5	Dec 2023
2	Describe the process of creating an inverted index. How can this be optimized with the help of block addressing?	10	Dec 2023
3	Describe the process of creating an inverted index with an example. How can this process be optimized using block addressing?	10	May 2023
4	Describe the process of creating an inverted index with an example. How can this process be optimized using block addressing?	10	May 2024
5	What is inverted Index? Describe the process of creating an inverted index without stop words for the following example	10	Dec 2024
6	What is inverted Index? Describe the process of creating an inverted index without stop words with example	10	Dec 2025
7	Explain Suffixes, Suffix Trie and Suffix tree, Suffix array and Supra Index with example	10	Dec 2024
8	Construct a Suffix, suffix trie and suffix tree, suffix array and supra index for the given sample text	10	May 2025
9	Discuss Huffman Algorithm in detail with suitable example	10	Dec 2024
10	Statistical text compression and Dictionary based text compression (short note)	5	May 2023
11	Sequential searching (short note)	5	Dec 2023

#	Question	Marks	Month & Year
12	Sequential Search (short note)	10	May 2024
13	Define Multimedia information retrieval. Discuss indexing and searching	10	Dec 2023
14	Define Multimedia information retrieval. Discuss Multimedia indexing and searching	10	Dec 2024
15	Multimedia indexing (short note)	10	May 2024
16	Explain Video Information Retrieval (VIR) and Image Information Retrieval (IIR) — 3 techniques each; how they improve user experience in multimedia search	10	Dec 2025
17	Automatic feature extraction (short note)	5	May 2023
18	Discuss concept of Text search engine	5	Dec 2023
19	Explain search engine Architecture	5	May 2023
20	What is the purpose of a search engine? List the steps of search operation by search engine	5	May 2024
21	Identify the various task performed by web search engine	5	May 2025
22	Describe meta searchers and its merits with example	5	Dec 2025
23	Web information retrieval system (short note)	5	May 2023

Module VI: User Interface and Visualization

#	Question	Marks	Month & Year
1	Discuss starting points. Explain list of collection and overviews in detail	10	Dec 2023
2	What is the starting point? Explain list of collections and overviews in detail	10	May 2024
3	Starting points (short note)	5	Dec 2024
4	Student result management system scenario: identify starting point of program, describe List and Collection interfaces, advantages of Collection Framework	10	Dec 2025
5	What is Query Specification? Describe direct manipulation and natural language in the context to Boolean query formulation	10	May 2023
6	What is human-computer interaction? List and discuss any four design principles for information access interfaces	10	May 2023
7	What is human-computer interaction? List and discuss any four design principles for information access interfaces	10	May 2024
8	Interface support for the search process (short note)	5	Dec 2023
9	Interface support for the search process (short note)	5	Dec 2024
10	Interface support for the search process (short note)	10	May 2024
11	Summarize two visualization techniques with respect to user interface design	10	May 2024
12	Summarize various visualization techniques with respect to user interface design	10	May 2025

#	Question	Marks	Month & Year
13	Recommend visualization techniques for a mobile banking app UX (account activity, financial trends)	5	Dec 2025
14	Importance of Context in Information retrieval (short note)	10	Dec 2025